Math 170E
Week 7

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1 Normal Random Variables

Example 1. (PSI 3.2.24) Let the random variable X be equal to the number of days that it takes a high-risk driver to have an accident. Assume that X has an exponential distribution. If $\mathbb{P}(X < 50) = 0.25$, compute $\mathbb{P}(X > 100|X > 50)$.

$$(A)\frac{1}{4} \qquad (B)\frac{1}{2} \qquad (C)\frac{\sqrt{2}}{2} \qquad (D)\frac{3}{4} \qquad (E)1-\frac{\sqrt{2}}{2}$$

Example 2. (PSI 3.3.1-modified) If Z is $N(0,1)$, find the following probabilities using standard normal table:

1. $\mathbb{P}(Z \leq 1.2)$
2. $\mathbb{P}(Z \leq 0.87)$
3. $\mathbb{P}(0.53 < Z \leq 2.06)$
4. $\mathbb{P}(Z \leq -1)$
5. $\mathbb{P}(-0.79 \leq Z < 1.52)$
6. $\mathbb{P}(Z > -1.77)$
7. $\mathbb{P}(Z > 2.89)$
8. $\mathbb{P}(|Z| < 1.96)$
9. $\mathbb{P}(|Z| < 2)$ (A)0.453 (B)0.671 (C)0.881 (D)0.954 (E)0.98

Example 3. (PSI 3.3.7) If X is $\mathcal{N}(3,4)$, find $\mathbb{P}(|X| < 4)$

(A)0.13 (B)0.25 (C)0.36 (D)0.56 (E)0.69

Example 4. (PSI 3.3.7) If X is $\mathcal{N}(650,400)$, find

1. $\mathbb{P}(600 \le X < 660) \approx$ (A)0.68 (B)0.72 (C)0.83 (D)0.94 (E)$None$
2. A constant $c > 0$ such that $\mathbb{P}(|X - 650| \le c) = 0.9544$
   (A)12.1 (B)24.7 (C)36 (D)39.2 (E)44.9

Example 5. (Old Quiz Problem) Let X be the standard normal $\mathbb{N}(0,1)$. Find the pdf of $Y = |X|$.

Example 6. (Old MT Problem) Given a constant c, suppose X is a continuous RV with pdf

$$f(x) = c(x+3), \qquad -2 < x < 1$$

1. Find c.
2. Let $Y = |X+1|$. Find the support and pdf of Y.

Example 7. (Old MT problem) Let us consider an unfair coin which tosses head and tail with probability 0.01 and 0.99, respectively.

1. Toss this coin independently, and let X be the first time that head appears. Compute the mean and variance of X.
2. Toss this coin for 500 times independently. Use a Poisson limit theorem to compute (approximately) the probability that exactly 5 many head appears.